Chapter Notes: Life Processes in Animals

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Introduction

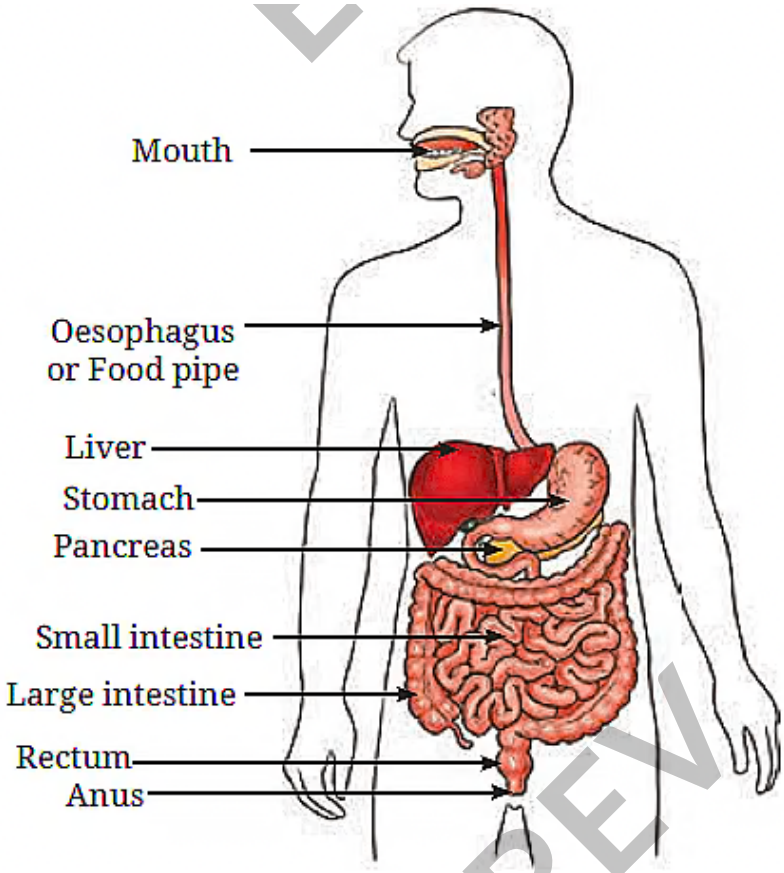
Animals eat different types of food depending on their habitat and habits. For example:

Bees and sunbirds suck nectar from flowers.
Human infants and the infants of many other animals feed on their mother's milk.
Snakes, such as pythons, swallow the animals they catch.
Some aquatic animals filter tiny food particles from water.

All animals, including humans, obtain energy from food. This energy is essential for carrying out life processes. Food contains complex components such as **carbohydrates, proteins, and fats**. These components must be broken down into simpler forms before the body can use them.

This breakdown of food takes place in a long tube called the **alimentary canal**, which starts at the **mouth** and ends at the **anus**. As food passes through this canal, **digestive juices** secreted by different organs help convert it into simpler substances. These nutrients are then **absorbed and transported** to various parts of the body for energy production, growth, repair, and other life processes.

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Human Digestive System

Nutrition in Animals

Animals obtain energy from food, but the way food is broken down and used varies among different species. To understand this process, let us study **digestion in humans**.

Digestion in Human Beings

> Digestion is the process of **breaking down complex** food into simpler substances. It occurs in the alimentary canal, a long tube that extends from the mouth to the anus.

The following stages describe how digestion takes place in human beings from beginning to end.

1. Beginning with the Mouth Cavity

Digestion begins in the **mouth**, where food enters and is broken down into smaller pieces by the **teeth** through **crushing and chewing**. This is called **mechanical digestion**.

When you think of your favourite food, your mouth produces more **saliva**. Saliva moistens the food and makes it easier to swallow.

Saliva contains the enzyme salivary amylase, which begins breaking down **starch**, a carbohydrate, into simpler sugars. This is why starchy foods such as chapati or rice may

taste sweet when chewed for some time.

Science and Society

A healthy mouth requires good oral hygiene. We should brush our teeth and clean our tongue twice a day. We should also rinse our mouth with water after each meal to help prevent tooth decay and bad breath. Find out the ways in which our elders maintained oral hygiene.

Activity to Investigate Saliva's Role

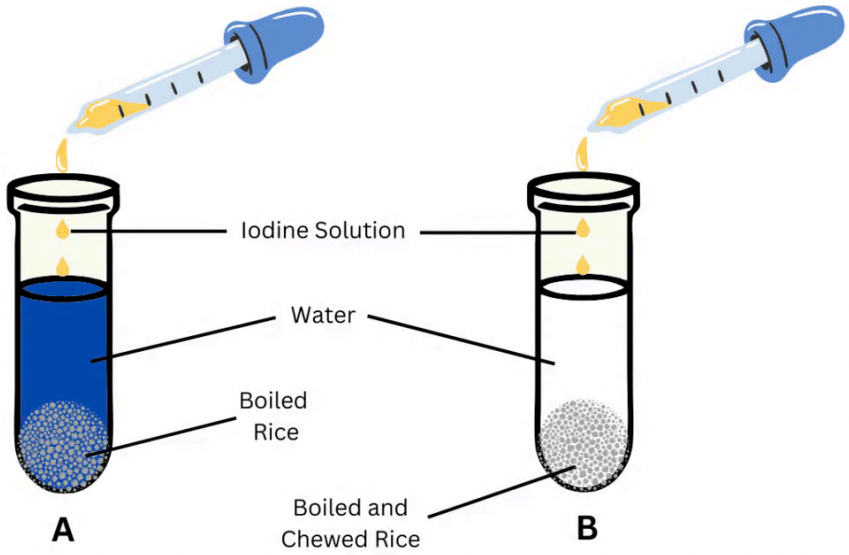
Aim: To observe the effect of saliva on the breakdown of starch in boiled rice.

Materials Needed: Two test tubes, a teaspoon, boiled rice, water, and iodine solution.

Procedure:

Label the test tubes as 'A' and 'B'.
Place one teaspoonful of boiled rice in test tube A and one teaspoonful of chewed boiled rice in test tube B.
Add 3–4 mL of water to both test tubes.
Note the initial colour of the rice-water mixture.
Add 3–4 drops of iodine solution to each test tube and mix the contents.
Observe the colour changes in both test tubes.

Observations:



In test tube A, the boiled rice may turn blue-black, indicating the presence of starch.

In test tube B, the chewed rice may not change colour or may show a very light blue-black colour. This indicates that some starch has been broken down into simpler sugars by saliva.

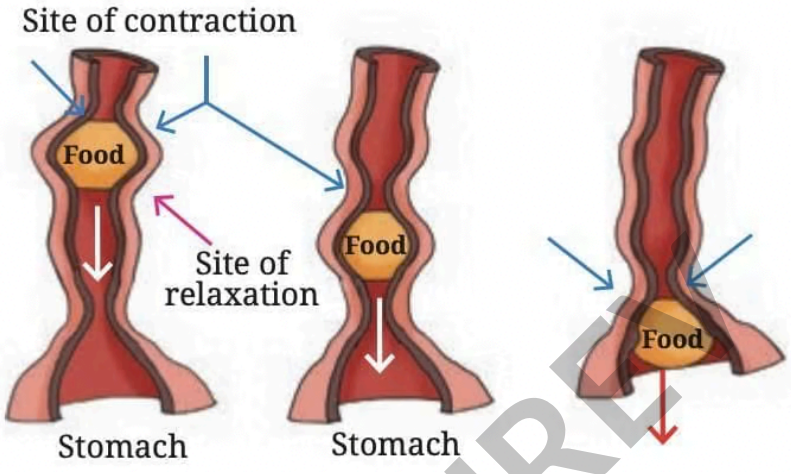
2. Food Pipe (Oesophagus): A Passage from the Mouth to the Stomach

After chewing, saliva moistens the food, making it soft and easy to swallow.

The tongue mixes the chewed food with saliva and pushes it into the **food pipe** or **oesophagus**.

The oesophagus is a long, flexible tube that connects the mouth to the stomach.

Food moves down the oesophagus by a wave-like movement called **peristalsis**. During peristalsis, the walls of the food pipe contract and relax gently.



Movement of food in the food pipe

This movement continues throughout the alimentary canal and pushes food forward for digestion.

3. Stomach

The walls of the food pipe contract and relax in a wave-like motion to push food down into the stomach.

The walls of the stomach contract and relax to **churn** the food and mix it thoroughly.

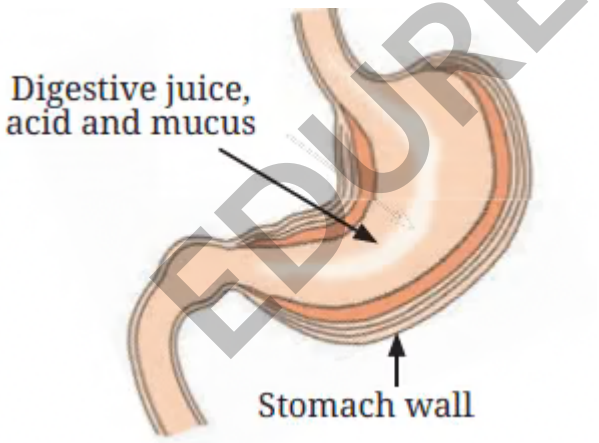
The stomach lining secretes **digestive juice**, **acid**, and **mucus**.

Digestive juice helps break down **proteins** into simpler substances.

The acid provides the acidic medium needed for protein digestion and helps kill harmful bacteria.

Mucus protects the stomach lining from being damaged by the acid.

Food is partially digested in the stomach and changed into a semi-liquid mass that is ready for the next stage of digestion.



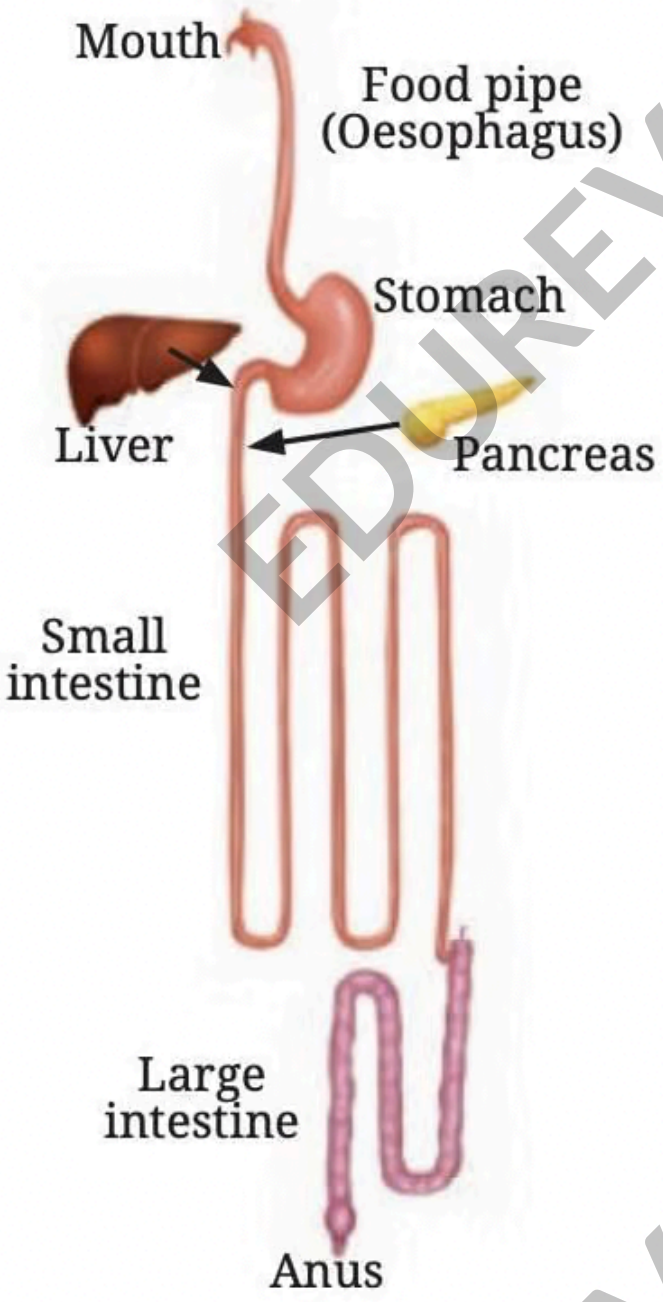
Fascinating Facts

In 1822, Alexis St. Martin was accidentally shot in the stomach. After treatment, a small permanent opening remained.
Dr William Beaumont, his doctor, used this opening to directly observe digestion in the stomach.
Beaumont conducted experiments to study how different foods are broken down.
He also explored how emotions can affect the process of digestion.
This accidental discovery significantly advanced our understanding of the functions of the stomach.

4. Small Intestine

After leaving the stomach, partially digested food enters the **small intestine**. It is about ***6 metres long***, making it the longest part of the alimentary canal.

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Alimentary canal if it is stretched out

The small intestine receives digestive secretions from:

1. Its own inner lining.
2. The **liver**, which produces **bile**.
3. The **pancreas**, which produces **pancreatic juice**.

Bile is mildly basic. It helps neutralise stomach acid and breaks fats into tiny droplets, making them easier to digest.

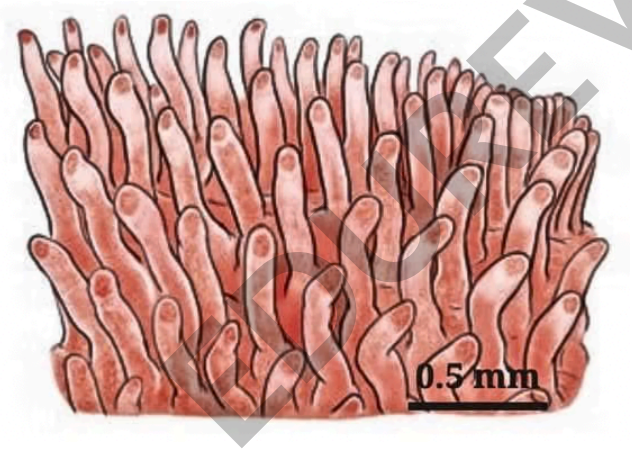
Pancreatic juice is also basic. It helps neutralise acid and contains substances that help break down carbohydrates, proteins, and fats.

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Digestive juices from the small intestine further break down fats, proteins, and carbohydrates into simpler substances.

Nutrients are absorbed into the blood through the **inner lining** of the small intestine.

The lining has thousands of finger-like projections called **villi**. These projections increase the surface area for efficient absorption.



Absorbed nutrients provide energy, help in growth and repair, and support the proper functioning of the body.

Science and Society

Celiac disease is a condition in which the body reacts to **gluten**, a protein found in wheat, barley, and rye.

This reaction damages the **inner lining of the small intestine** and affects nutrient absorption.

As a result, the small intestine cannot function properly.

The main way to manage celiac disease is to **avoid gluten-containing foods**, as advised by a healthcare professional.

Millets such as jowar, bajra, and ragi are good alternatives because they are naturally **gluten-free**.

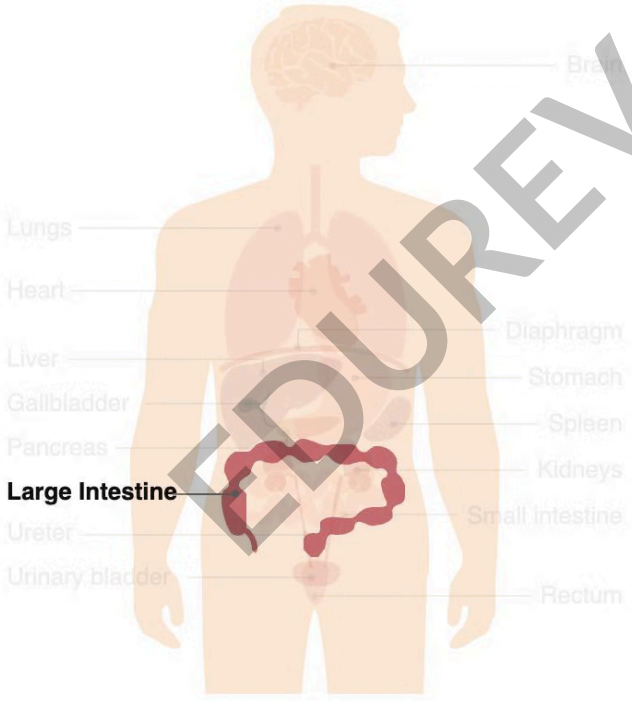


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5. Large Intestine

After nutrient absorption in the small intestine, undigested food moves into the **large intestine**.

The large intestine is about **1.5 metres long**. It is shorter but wider than the small intestine.



Its main function is to **absorb water and some salts** from undigested food.

This process changes the waste into a **semi-solid form called stool**.

Stool is stored in the **rectum** until the body is ready to eliminate it.

The waste is expelled through the **anus** in a process called **egestion**.

Eating **fibre-rich foods**, such as fruits, vegetables, and whole grains, helps keep the large intestine healthy and makes stool easier to pass.

Fascinating Facts

> The large intestine contains helpful bacteria that break down some undigested fibre.
> These bacteria produce substances that are important for health.
> Eating fibre-rich and fermented foods such as curd and pickles can support healthy digestion.

Science and Society

EDUREV

The ancient Ayurvedic text *Charaka Samhita* stresses the importance of easily digestible foods.

Spices such as ginger, black pepper, and cumin are traditionally used to support digestion.

Modern nutrition also highlights eating at proper times, eating mindfully, and avoiding overeating to maintain good digestive health.



Digestion in Other Animals

Not all animals digest food in the same way as humans. Their digestive systems are adapted to their diets and habitats.

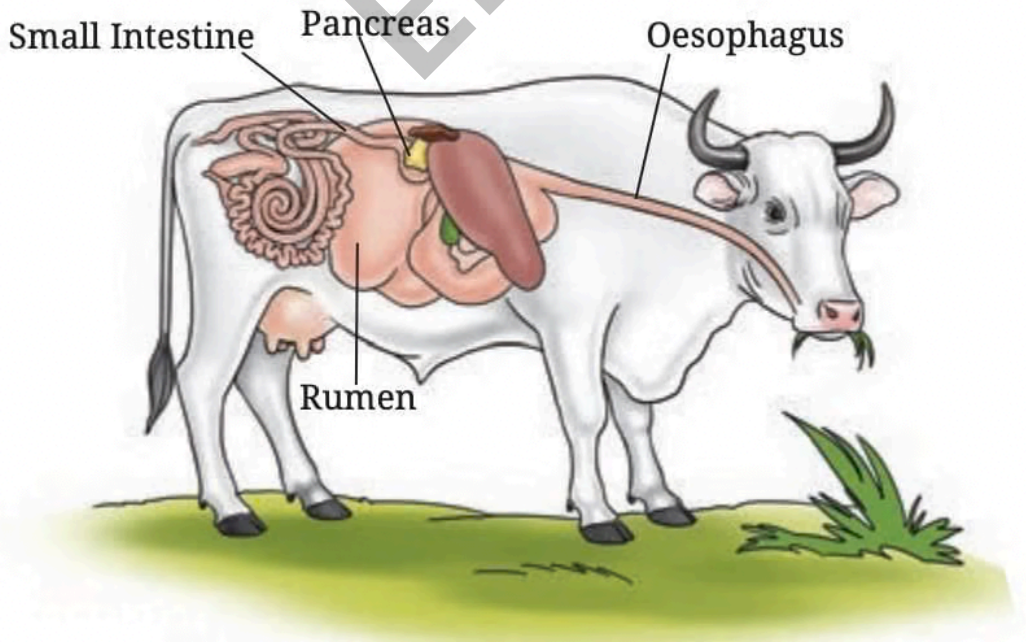
1. Ruminants

Ruminants are grass-eating animals. They partially chew grass and swallow it into their stomachs, where it undergoes partial digestion.

The food is then brought back to the mouth for thorough chewing. This process is called **rumination**.

Ruminants may spend about 8 hours a day chewing their food.

After thorough chewing, the food moves down the alimentary canal for further digestion.



2. Birds

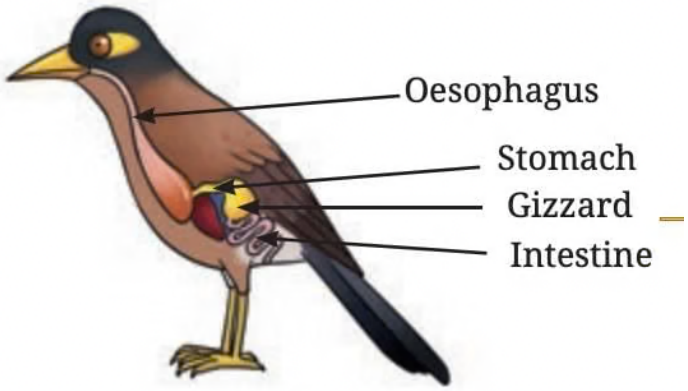
Birds lack teeth but have a special muscular stomach chamber called the **gizzard**.

The gizzard breaks down food by contracting and relaxing. Small stones, called **grit**, swallowed by birds may help grind the food.

Animals show variations in their digestive systems according to the types of food they eat.

Nutrients from digested food help build and repair the body or are broken down to release energy.

The process of using nutrients to release energy is called **respiration**.



Respiration in Animals

Respiration is the process by which animals use oxygen to break down nutrients such as glucose and release energy. Carbon dioxide and water are formed as products of this process.

Breathing is a physical process that brings oxygen into the body and removes carbon dioxide. Respiration is a chemical process that takes place in cells.

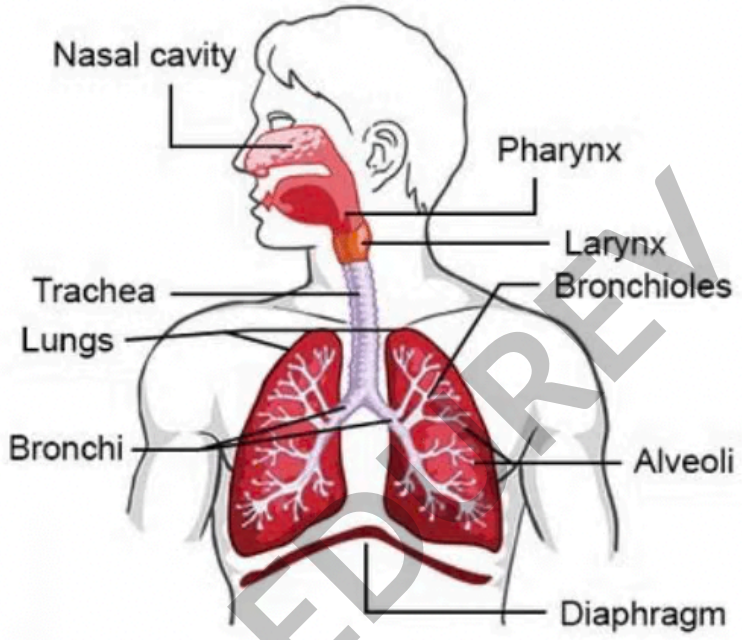
Respiration in Humans

Breathing is the process of inhaling, or breathing in, oxygen-rich air and exhaling, or breathing out, air containing more carbon dioxide.

Breathing is essential for survival. Without breathing, humans cannot live for more than a few minutes.

All living organisms exchange gases with their surroundings. Plants and animals also carry out respiration.

The **respiratory system** is the body system responsible for breathing and gas exchange.



Human Respiratory System

Nostrils: Openings in the nose through which air is inhaled and exhaled.

Nasal passages: Passages after the nostrils that are lined with tiny hairs and mucus. They trap dust and dirt. Breathing through the nose is better than breathing through the mouth because of this filtration.

Windpipe (trachea): A tube that carries air from the nasal passages to the lungs and divides into two branches.

Lungs and alveoli: The windpipe branches further inside the lungs into finer tubes that end in tiny balloon-like sacs called **alveoli**, where gas exchange occurs.

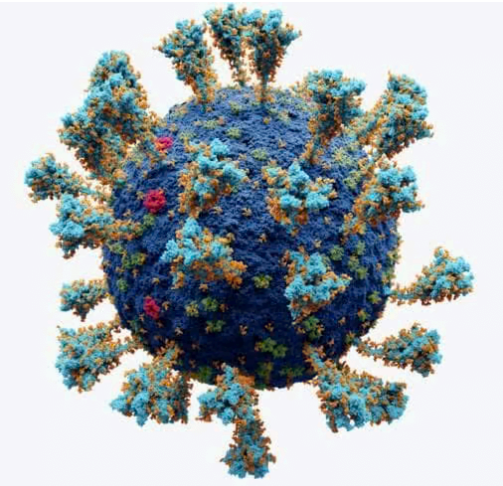
Protection: The lungs are protected by the rib cage, a bony structure surrounding them.

Science and Society

Although the respiratory system filters out much of the dust in inhaled air, small infectious particles can still enter the lungs.

For example, during the COVID-19 pandemic, the **SARS-CoV-2 virus** affected the respiratory system.

This virus caused breathing difficulties and serious lung problems in many people.



Activity: Understanding the Breathing Mechanism through a Simple Model

Materials Needed: A wide transparent plastic bottle with a lid and its bottom removed, a Y-shaped hollow tube, two deflated balloons, rubber bands, clay for sealing, a thin rubber sheet, and a large rubber band for securing the rubber sheet.

Procedure

Prepare the Bottle:

Take the wide transparent plastic bottle and remove its bottom.

Make a hole in the lid of the bottle.

Prepare the Tube and Balloons:

Take the Y-shaped hollow tube. Fix the two deflated balloons to the forked ends of the tube and secure them with rubber bands to make them airtight.

Assemble the Model:

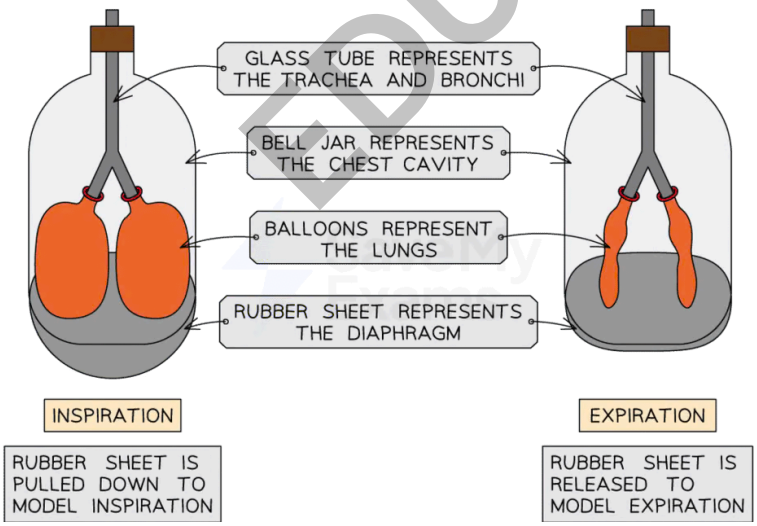
Insert the straight end of the tube tightly through the lid of the bottle and seal the lid with clay to make it airtight.

Attach a thin rubber sheet tightly to the open base of the bottle using a large rubber band.

Conducting the Experiment:

Pulling the rubber sheet downward: Pull the rubber sheet from the centre of the base downwards. Observe the balloons closely.

Releasing the rubber sheet: Release the rubber sheet upwards and observe the changes in the balloons.



Model of Mechanism of Breathing

Observations:

When you pull the rubber sheet downwards, the balloons inflate.

When you release the rubber sheet upwards, the balloons deflate.

How Respiration Happens

(i) Inhalation

When you **breathe in**, or inhale, your **chest expands** as your **ribs** move up and outwards.

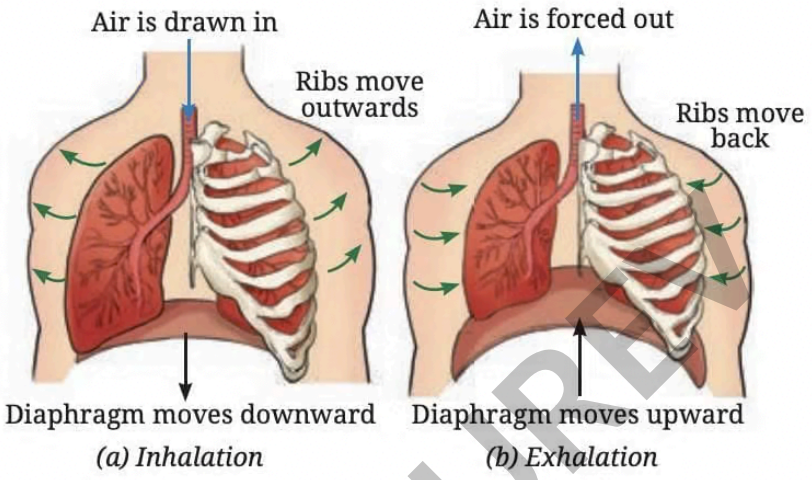
The **diaphragm** is a dome-shaped muscle located below the lungs. It moves downwards during inhalation.

This movement increases the space inside the chest and allows **air to enter** the lungs.

(ii) Exhalation

When you **breathe out**, or exhale, your ribs move down and inwards, while the diaphragm moves upwards.

This reduces the space inside the chest and helps to **push air out** of the lungs.



In the activity described above, the balloons represent the lungs and the rubber sheet represents the diaphragm.

Science and Society

Breathing exercises such as Pranayama and Tummo have been practised for centuries in different parts of the world to support calmness and well-being. **Pranayama** from India involves controlled breathing and may support concentration. **Tummo** breathing, associated with cold regions such as Ladakh, is traditionally practised to help develop body warmth. Deep breathing combined with chanting is also used in many traditions to promote relaxation and mental clarity.



What Do We Breathe Out?

When we exhale, the air we breathe out contains more carbon dioxide than the air we inhale.

This can be demonstrated through an experiment with lime water.

Lime water is a clear solution that turns milky when it reacts with carbon dioxide.

Two test tubes containing equal amounts of fresh lime water are taken.

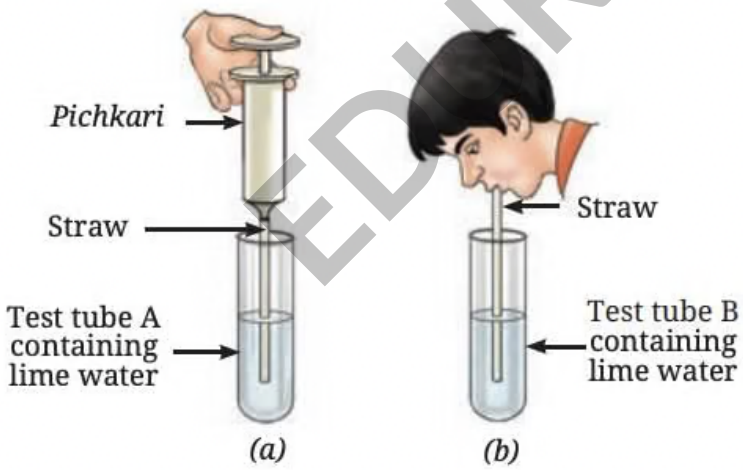
Air similar to inhaled air is passed through the lime water in test tube A using a syringe.

Exhaled air is blown through a straw into the lime water in test tube B.

Test tube B, through which exhaled air is passed, turns milky because of its reaction with carbon dioxide.

Test tube A, through which air similar to inhaled air is passed, shows little or no change and remains clear.

This indicates that exhaled air contains a higher concentration of carbon dioxide than inhaled air.



(a) Air is passed into lime water with a pichkari or syringe; (b) air is exhaled into lime water

(iii) Gas Exchange

Fresh air enters the lungs during breathing and fills tiny sacs called **alveoli**.

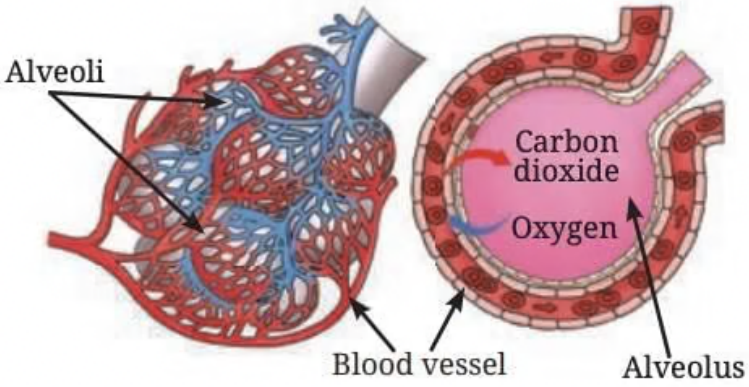
Alveoli have thin walls and are surrounded by blood vessels.

Carbon dioxide from the blood passes into the alveoli and is removed from the body during exhalation.

Oxygen from the alveoli passes into the blood and is carried to the entire body.

Oxygen is used to break down glucose from food and release energy in a process called **respiration**.

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Gas exchange through alveoli

Overall Process of Respiration: In cells, oxygen breaks down glucose to release energy, carbon dioxide, and water.

$$Glucose + Oxygen \rightarrow Carbon\ dioxide + Water + Energy$$

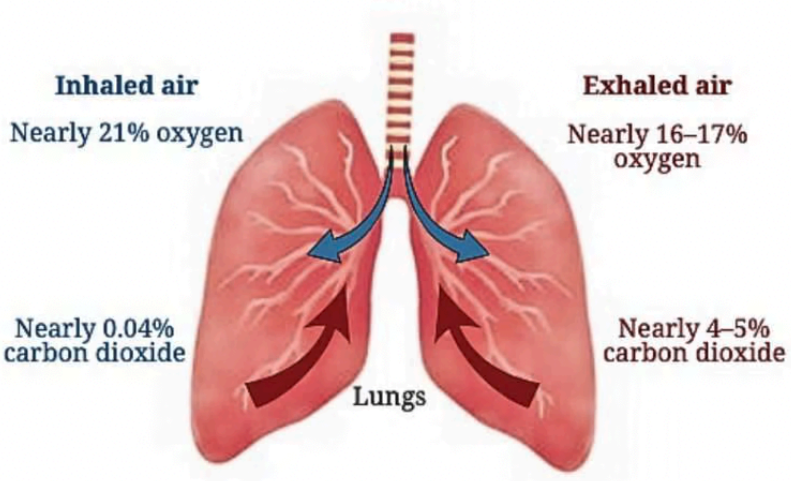
Inhaled air contains about **21% oxygen and 0.04% carbon dioxide**.

Exhaled air contains about **16–17% oxygen and 4–5% carbon dioxide**.

Breathing is a **physical process** involving the intake and release of air.

Respiration is a **chemical process** inside cells that produces energy.

Both breathing and respiration are vital for survival.



The percentage of oxygen and carbon dioxide in inhaled and exhaled air

Difference Between Breathing and Respiration

Breathing and respiration are related but occur at different levels in the body.

Breathing is the simple act of moving air in and out of the lungs, whereas respiration is the chemical process inside cells that releases energy from food.

The following visual compares these two processes.

EDUREV

Feature	Breathing	Respiration
Type of process	Physical process	Chemical process
Involves	Inhaling oxygen and exhaling carbon dioxide	Breaking down food (glucose) using oxygen to release energy
Occurs in	Lungs and air passages	Cells throughout the body
Purpose	To bring in oxygen and remove carbon dioxide	To produce energy for body functions
Requires energy?	No	Yes (it releases energy)
Produces	Only exchange of gases	Energy (ATP), carbon dioxide, and water
Control	Can be voluntary or involuntary	Completely involuntary

EDUREV

Role of the Circulatory System

The **circulatory system** transports nutrients, oxygen, and waste products throughout the body.

It consists of the **heart**, **blood**, and **blood vessels**.

The heart pumps blood through blood vessels to deliver oxygen and nutrients and remove waste products.

Science and Society

Smoking damages the lungs and increases the risk of lung cancer and other respiratory diseases.

It causes **persistent coughing** and makes the body more prone to frequent infections.

Smoking releases **toxic chemicals** into the air and harms other people through **passive smoking**.

Children, pregnant women, and older people are especially vulnerable to passive smoking.

Avoiding smoking protects both personal health and the health of people around us.

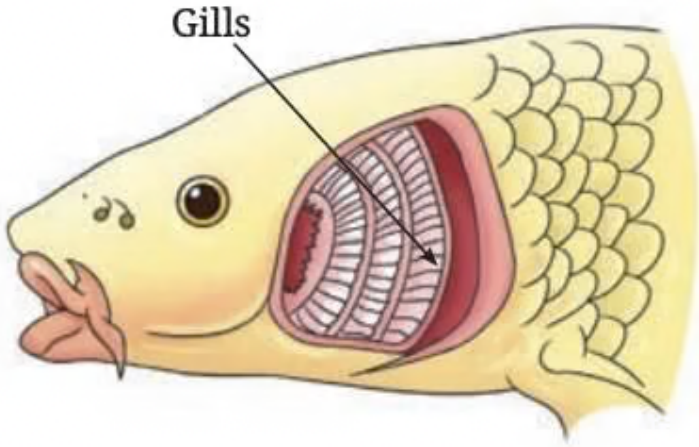
Respiration in Other Animals

Different animals live in different habitats and have **different breathing mechanisms** suited to their surroundings.

Animals have respiratory systems adapted to their habitats:

Lungs: Animals such as birds, elephants, lions, cows, lizards, and snakes breathe through lungs, although the structure of their lungs may vary.

Gills: Aquatic animals such as fish use gills to extract dissolved oxygen from water and release carbon dioxide.



Breathing body parts in a fish

Skin and gills: Amphibians such as frogs use gills as tadpoles, lungs as adults on land, and moist skin for gas exchange in water.

Moist skin: Earthworms breathe through their moist skin, which allows the exchange of oxygen and carbon dioxide.

Terms to Remember

Alimentary canal: The long tube in the digestive system that extends from the mouth to the anus and in which food is digested and absorbed.

Mechanical digestion: The physical breakdown of food into smaller pieces, such as by chewing.

Peristalsis: Wave-like muscle contractions that move food through the digestive system.

Villi: Finger-like projections in the small intestine that increase the surface area for nutrient absorption.

Egestion: The process of expelling undigested waste, or stool, through the anus.

Ruminants: Animals such as cows that partially digest food, bring it back to the mouth, and chew it again.

Gizzard: A muscular chamber in birds that grinds food, often with the help of swallowed stones.

Alveoli: Tiny air sacs in the lungs where oxygen and carbon dioxide are exchanged.

Diaphragm: A dome-shaped muscle below the lungs that helps in breathing by moving up and down.

Respiration: The chemical process in cells in which oxygen helps break down glucose to release energy.

Circulatory system: The system made up of the heart, blood, and blood vessels that transports nutrients, oxygen, and waste products.

You can practise questions from this chapter here: **Short & Long Question Answers: Life Processes in Animals** (https://edurev.in/studytube/Class-7-Science-Chapter-9-Question-Answers--Life-Processes-in-Animals/b9cb9bd6-ee36-41c8-b456-86f80e55532a_t)